

$$P(x)=2x^3-48x^2+256x+8$$

$$\text{Find } \int_0^{16} P^{1729}(x)dx$$

$$\text{where } P^n(x)=P(P^{n-1}(x))\forall n\geq 2\wedge P^1(x)=P(x)$$

$$u:=P^{1729}(x)$$

$$u=P\Big(P\Big(P\Big(\overset{\dots}{\underset{1726}{(x)}}\Big)\Big)\Big)$$

$$P(x)=2(x^3-24x^2+128x+4)$$

$$P(x)=2(x(x^2-24x+128)+4)$$

$$P(x)=2(x(x-8)(x-16)+4)$$